

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 1: Information Representation

Subtopic 1.3: Compression

Concept 1.3.1: Need for Compression

Total Questions: 7

Mark Schemes begin on Page 5

Question 1 | Source: 9618_s25_qp_11 (Q2.b.i)

- 2 Programmers in a software development company take part in live video conferences to discuss their work.

The live video conferences take place using real-time bit streaming. The video is compressed before it is transmitted.

- (b) (i) Explain the reasons why a video is compressed before it is transmitted using real-time bit streaming.

.....

.....

.....

.....

.....

.....

.....

..... [4]

Question 2 | Source: 9618_w24_qp_11 (Q4.b.i)

- 4 A student uses a laptop to write a program that is saved as a text file.

- (b) The student compresses the file before it is emailed to their teacher as an attachment.

- (i) Explain the benefits to the teacher of the attachment being a compressed file.

.....

.....

.....

.....

.....

..... [3]

Question 3 | Source: 9618_s24_qp_12 (Q2.d.iii)

- 2 A computer game is being designed that users will be able to play using a virtual reality (VR) headset.

- (d) The computer can transmit a video made from bitmap images and vector graphic animations to the VR headset.

- (iii) The bitmap video is **not** compressed before transmission to the VR headset.

Give **two** reasons why the video does **not** need to be compressed.

- 1
- 2
- [2]

Question 4 | Source: 9618_w23_qp_11 (Q1.c)

- (c) Explain why a bitmap image is often compressed before it is attached to an email.

.....

.....

.....

..... [2]

Question 5 | Source: 9618_w23_qp_13 (Q5.b.i)

- 5 A programmer uses an Integrated Development Environment (IDE) to develop a program that monitors air quality.

- (b) The program is distributed by downloading the source code and its library files from a web server.

- (i) Explain the reasons for compressing the files.

.....

.....

.....

..... [2]

Question 6 | Source: 9618_s23_qp_11 (Q1.c.i)

- 1 Images are being created to advertise holidays.

Some of the images are bitmap images and some are vector graphics.

- (c) The photographs are compressed before they are uploaded to a web server. Customers download the photographs from this web server.

- (i) Explain the reasons why compressing the photographs will benefit the customers.

.....

.....

.....

.....

.....

..... [3]

Question 7 | Source: 9618_w21_qp_11 (Q7.b.i)

- 7 Bobby is recording a sound file for his school project.

- (b) Bobby wants to email the sound file to his school email address. He compresses the file before sending the email.

- (i) Explain the reasons why Bobby compresses the sound file.

.....

.....

.....

..... [2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_s25_ms_11 (Q2.b.i)

2(b)(i)	1 mark each to max 4 e.g. • Video is data-intensive • The file size needs reducing in order to reduce the amount of bandwidth used and reduce buffering • This means people are not behind in the conversation and people with lower bandwidth can still take part	4
---------	---	----------

Mark Scheme for Question 2 | Source: 9618_w24_ms_11 (Q4.b.i)

4(b)(i)	1 mark for each bullet point (max 3) e.g. • Less of the teacher's storage space is used • ... so more student work can be stored • Transmission time is reduced • ... so the teacher does not have to wait as long for it to arrive • Bandwidth usage is reduced • ... so other transmissions are not adversely affected • Less data allowance is used on the teacher's email system	3
---------	---	----------

Mark Scheme for Question 3 | Source: 9618_s24_ms_12 (Q2.d.iii)

2(d)(iii)	1 mark each to max 2 : • Dedicated connection to the headset // not sharing bandwidth • Already fast connection that can transmit the data without slowing • Video may already be a small file size and does not need further reduction • Video is not saved so storage is not an issue in the headset	2
-----------	---	----------

Mark Scheme for Question 4 | Source: 9618_w23_ms_11 (Q1.c)

1(c)	1 mark for each bullet point (max 2) • Reduced bandwidth usage when transmitting the message • Reduced transmission time from email client to email server • Reduced storage space on the email • Email accounts often have a maximum size for an attachment	2
------	--	----------

Mark Scheme for Question 5 | Source: 9618_w23_ms_13 (Q5.b.i)

5(b)(i)	1 mark for each bullet point (max 2). • To reduce the time it takes to download the program files from the web server // to upload them to the server in the first place • To reduce the amount of storage space used on the web server // user's device	2
---------	--	----------

Mark Scheme for Question 6 | Source: 9618_s23_ms_11 (Q1.c.i)

1(c)(i)	1 mark each Examples: • The customers will be able to download the photographs in less time • ...and they will take less of the customer's bandwidth • The photographs will take up less space on the customer's storage medium • ...therefore the customers can store more images • ...and will have more space for other files	3
---------	--	----------

Mark Scheme for Question 7 | Source: 9618_w21_ms_11 (Q7.b.i)

7(b)(i)	1 mark per bullet point • Reduces the file size • Faster to transmit/download • Original file is too large for email storage/attachment	2
---------	--	----------